D.3 LLM as a Judge Prompt

LLM as a Judge Prompt

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Please evaluate these two conflicting responses and determine which one is correct. Your response should:
1. Analyze both reasoning paths
2. Provide your final conclusion and answer

Your output should end with:
ANSWER: [letter]

Where [letter] is your final answer.

Question: {question_text}

First Response: {response_1}

Second Response: {response_2}
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E Quality Scoring Prompt

Quality Scoring Prompt

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You are an expert evaluator specializing in logic, argumentation, and critical thinking. Your task is to analyze the following multiple choice question and the provided solution and score the quality of the solution's reasoning based on the detailed rubric provided below.
Your analysis must be objective and rigorous. You will provide an overall score at the end.

### Evaluation Rubric

You will score the text on a scale of 1 to 5 for each of the five categories below.

1: Poor - The criterion is almost entirely unmet.

2: Weak - The criterion is met in a minimal or flawed way.

3: Average - The criterion is met, but with notable weaknesses or inconsistencies.

4: Good - The criterion is well-met, with only minor issues.

5: Excellent - The criterion is met flawlessly and effectively.

#### Category 1: Clarity & Precision

Focus: How clearly and precisely is the argument articulated?

5 (Excellent): The language is exceptionally clear, specific, and unambiguous. Key terms are explicitly defined and used consistently.

3 (Average): The argument is generally understandable, but contains some ambiguous phrases, imprecise language, or undefined key terms.

1 (Poor): The argument is vague, convoluted, and difficult to follow. It relies on jargon or ambiguous language that obscures the meaning.

#### Category 2: Premise Plausibility & Soundness

Focus: How true, plausible, and well-founded are the core premises or assumptions upon which the argument is built?

5 (Excellent): The core premises are demonstrably true or highly plausible and are widely accepted or well-defended. The argument rests on a solid foundation.
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3 (Average): The premises are plausible but debatable, or they are a mix of strong and weak assumptions. The foundation has some potential weaknesses.

1 (Poor): The core premises are demonstrably false, highly implausible, or based on baseless assumptions. The entire argument is built on a faulty foundation.

#### Category 3: Logical Coherence

Focus: Does the argument follow a logical progression? Are the conclusions well-supported by the premises?

5 (Excellent): The reasoning is flawlessly logical. Conclusions follow irrefutably from the premises. The structure is sound, and there are no logical fallacies.

3 (Average): The main line of reasoning is logical, but there are some gaps, inconsistencies, or minor fallacies that weaken the argument.

1 (Poor): The argument is illogical, inconsistent, or riddled with significant logical fallacies (e.g., ad hominem, straw man). The conclusion does not follow from the premises.

#### Category 4: Evidence & Factual Grounding

Focus: Are the claims supported by credible, relevant, and sufficient evidence?

5 (Excellent): All key claims are supported by strong, credible, and directly relevant evidence that is accurately interpreted.

3 (Average): The argument presents evidence, but it may be of mixed quality, tangential, misinterpreted, or based on limited data.

1 (Poor): Claims are largely unsupported, based on anecdote, opinion, or unreliable sources. Evidence is absent, irrelevant, or factually incorrect.

#### Category 5: Depth & Nuance

Focus: Does the reasoning engage with the complexity of the topic?

5 (Excellent): The reasoning is sophisticated and nuanced. It thoughtfully considers and addresses potential counterarguments, acknowledges limitations, and explores underlying assumptions.

3 (Average): The reasoning shows some consideration of complexity but tends to be one-sided, mentioning alternative viewpoints without engaging them meaningfully.

1 (Poor): The reasoning is simplistic and one-dimensional, ignoring or dismissing counterarguments and complexity.

### Question and Solution to Evaluate
{Question}

Solution: {Solution}

#### Required Output Format

Please end your output with a list containing scores for each of the five categories

Scores: [Category 1 Score, Category 2 Score, Category 3 Score, Category 4 Score, Category 5 Score]

example output: [5, 3, 4, 2, 5]
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F Persuasion Probability Aggregated by Dataset

Table 11: Probabilities Aggregated by MCQ Dataset

Dataset	N	$P(T(R_i))$	$P(T(R_f))$	$\mathcal{F}$	$\mathcal{F}_c$	$\mathcal{F}_i$	$P(T(R_i) \mid R_f = R_r)$	$P(\neg T(R_i) \mid R_f = R_r)$	$P(R_f \neq R_i \wedge R_f \neq R_r)$
LogiQA	6720	48.0%	55.6%	54.1%	47.8%	60.2%	40.5%	59.5%	1.5%
MedMCQA	2916	48.1%	59.4%	47.3%	37.0%	57.2%	34.5%	63.4%	0.9%
MMLU	2664	49.1%	62.4%	45.2%	31.4%	58.8%	27.7%	70.2%	2.0%
MMLU-Pro	4746	48.1%	60.4%	50.1%	37.2%	62.2%	30.9%	69.1%	3.4%
CommonsenseQA	4368	47.9%	55.0%	61.9%	56.8%	66.8%	43.7%	56.3%	1.0%

Symbol Definitions:

- N: Total count of disagreement pairs.
- R_i: Initial response.
- R_f: Final response.
- R_r: Refuting response.
- $T(R_x)$ / $\neg T(R_x)$: Event R_x is true/false.
- $\mathcal{F}$: $100 \cdot P(R_f = R_r)$.
- $\mathcal{F}_c$: $100 \cdot P(R_f = R_r \mid T(R_i))$.
- $\mathcal{F}_i$: $100 \cdot P(R_f = R_r \mid \neg T(R_i))$.

Further Column Context (as %):

- $P(T(R_i) \mid R_f = R_r)$: Prob. R_i correct given the model was persuaded.
- $P(\neg T(R_i) \mid R_f = R_r)$: Prob. R_i incorrect given the model was persuaded.
- $P(R_f \neq R_i \wedge R_f \neq R_r)$: Prob. R_f is a new answer.